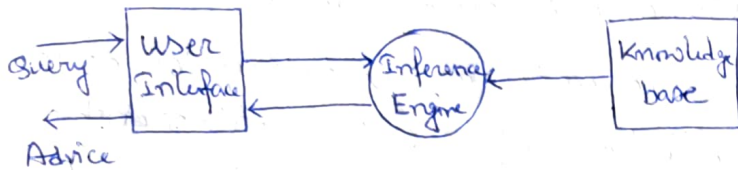


Expert System



Expert system is able to do the work of a professional.

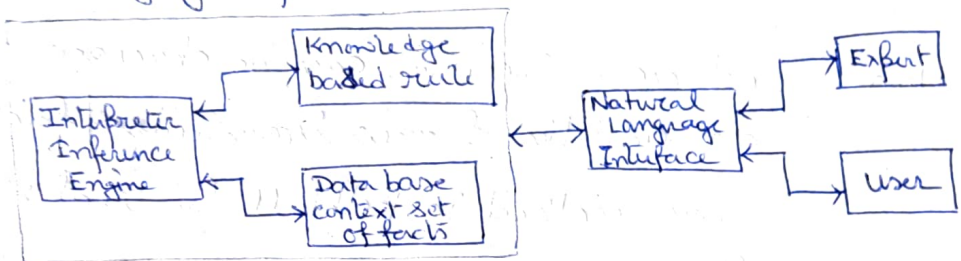
Characteristics -

i) High performance

ii) Reliable

iii) Highly Responsive

iv) Understandable



Knowledgebase - all data and ~~is~~ information are input by experts.

Store data as set of rules that the system must follow to make decision

Inference Engine - ask user for query
applies rules and knowledge from knowledge base
appropriately uses knowledge to reach to a decision

Knowledge base ~~can be~~ reasoning can be of two types — Monotonic and Non-monotonic reasoning.

Monotonic — once decision is taken based of some information, that will not be further changed although new information is added. Hence this method is not appropriate for realtime system.

Advantage: all old proofs are valid.

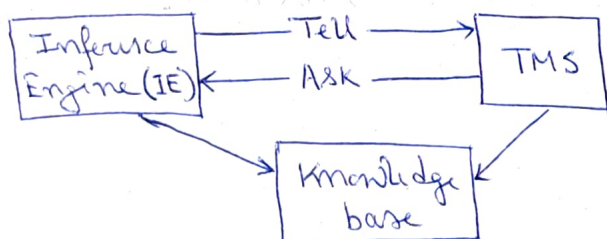
Non-monotonic — Decision can be changed based on addition of new knowledge/fact.

This method is helpful for real world scenario.

Deals with problems with incomplete and uncertain models.

Truth Maintenance System (TMS)

TMS implements to permit a form of non-monotonic reasoning by permitting the addition of changing statements to knowledge base.



Problem Solver

TMS helps to solve problem. It supports backtracking to recover from inconsistency.